

Response to Reviewers

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At the Point of Entry: Receiving Help and Newcomer Contribution on Stack Overflow

We thank the editor and both reviewers for their careful reading. In preparing this revision we re-matched treated and control questions — nearest-neighbor propensity-score matching *with replacement*, caliper 0.05, exact strata on calendar year, primary tag, and number of tags — and re-fit every Cox model, so all reported estimates derive from a single pipeline run. The principal changes are matched-pair bootstrap inference, a decomposition of the helping outcome by help type, a non-parametric response-time specification re-estimated with answer-quality controls, new robustness checks for observable selection and entry cohort, a shortened Theory section, and a re-structured Discussion. One transparency item: a legacy appendix analysis of lifetime contributions was removed because it derived from a superseded sample and constructs not defined in the current pipeline; no claim in the revised manuscript relies on it. Replication code and instructions are in the package described in the Data availability statement.

To keep this letter short enough to be useful, we have followed one rule throughout. Where we accept a point and made the change, we say so in a sentence or two and point to the revised text. Where we did something different from what was asked, or where we disagree, we give the reasoning in full.

Editor requirements

- **Point-by-point response:** attached (this document), covering every Reviewer 1 sub-point (Q1–Q9), every Reviewer 2 point (1–9), and each editor requirement.
- **Editable source files:** the revised manuscript is uploaded in editable source form alongside the PDF, per the production requirement.
- **Highlights:** a separate **Highlights** file accompanies the submission with four bullet points, each verified at or below 85 characters including spaces, worded consistently with the title and the revised abstract.
- **Graphical Abstract:** we respectfully decline the optional Graphical Abstract.

Response to Reviewer 1

Reviewer comment	Response and changes to the manuscript
(Q1) “The authors should separate the empirical question from the proposed psychological mechanism. The data can show whether users answer more after receiving help, but they do not directly measure gratitude, moral obligation, or incentive displacement.” (Q5) “The ‘re-engagement window’ interpretation is also plausible, but the paper does not observe sessions, exits, returns, or notification behavior. This should be presented as a possible explanation rather than as an established mechanism.” (Q7) “The authors should acknowledge that they do not directly observe psychological mechanisms such as gratitude, obligation, or status motivation.”	Agreed, and revised throughout. Gratitude, moral obligation, social expectation, incentive displacement, and re-engagement are now marked as <i>proposed</i>, unmeasured interpretations in the abstract, the Theory section, and Sections 6.3 and 6.5. Social expectation is a third proposed pathway added to the Theory section in response to Reviewer 2’s P8, and Section 6.5 states that our design cannot adjudicate among them and leaves that to research with the instruments to measure psychological states directly. No models were re-fit for this change.

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(Q1) <i>“The objectives are generally clear, but the rationale could be stated more modestly. The paper should clarify that its main contribution is not to establish a general theory of generalized reciprocity, but to test a short-run behavioral response to receiving an answer on Stack Overflow.”</i> (Q6) <i>“The tenure result is potentially the most interesting contribution and should receive more emphasis than the broader claims about moral reciprocity.”</i> (Q5) <i>“The conclusion should emphasize the narrower finding: receiving an answer is associated with a small short-run increase in helping, mainly among newer users.”</i>	Agreed. The tenure gradient is now the headline in the abstract, the contributions paragraph, and the Conclusion (Section 7), which states that the response “is strongest among newcomers and fades to nothing among the longest-tenured users.” We also retitled the paper from “Help Converts Newcomers, Not Veterans” to “At the Point of Entry,” replacing the causal verb with associational language that matches the design.

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(Q2) <i>“The authors should clearly distinguish questions, matched observations, interval rows, users, and help events. The reported sample sizes and event counts in the main tables are difficult to reconcile.”</i> <i>“The paper should explain more clearly how controls are duplicated or reused in matching, and how repeated observations from the same user are handled.”</i> (Q4) <i>“The paper should include clearer reporting of the estimation sample in each table: number of users, questions, matched observations, interval rows, and events where relevant.”</i>	Agreed. The Data subsection is now an explicit filtering-and-matching chain, and new Appendix Table E.11 separates the five quantities per model, pooled and by tenure bucket. Two clarifications the reviewer’s question exposed, both now stated in the paper. Controls are reused: the matched treated questions draw on 2,481,556 <i>distinct</i> controls, about 4.9 times each. And estimation rounds interval boundaries to a one-hour grid, so same-hour events merge — this is what separates the descriptive event mean from the estimation Events counts (“Implementation Details”).

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Reviewer comment	Response and changes to the manuscript
(Q3) “Standard errors are a concern. The appendix states that robust or clustered standard errors are not used. Given repeated observations by users, repeated controls, matched pairs, and recurrent helping events, the current inference may be too optimistic.” “The authors should report robust standard errors clustered at least by user or matched pair, or use a bootstrap that preserves the matched structure.”	We implemented the second of the two remedies offered, because the first was not available to us: <code>lifelines</code>’ <code>CoxTimeVaryingFitter</code> raises <code>NotImplementedError</code> for <code>robust=True</code> and exposes no cluster column for time-varying fits. Resampling whole matched pairs and refitting Model A gives a pooled 95% CI of [1.032, 1.075] (HR 1.049; Table F.12). Because the concern bites hardest where our claims are strongest, we ran the newcomer stratum at 500 replicates with full-stratum refits: the bootstrap interval [1.057, 1.084] <i>confirms</i> the model-based interval rather than widening it. One limit, since the reviewer names repeated controls specifically: resampling pairs does not absorb the dependence induced by control <i>reuse</i>. We judge it second-order — the pair bootstrap already widens the pooled interval roughly fourfold over the model-based standard error — and present these intervals as bootstrap-based rather than fully cluster-robust. We also deleted the assertion that clustering was unnecessary, and now state that Cox-table standard errors are model-based and can be too narrow. Relatedly, we removed the summed contrast $\exp(\beta_2 + \beta_4)$ from the paper: β_2 carries no sign guarantee, so the sum bounds nothing. It is now reported as an explicit parallel-trends diagnostic with its own standard error.

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(Q3) “Matching on activity history and tag-level answer rates helps, but the authors should add stronger robustness checks, such as controls for posting hour/day, question length, number of tags, user reputation, and text-based measures of question quality or complexity.” (Q7) “The paper should more clearly acknowledge that answer receipt is not random and may reflect unobserved question quality or user characteristics.”	We added every requested covariate except one: posting hour-of-day and day-of-week, question body and title length, and owner reputation now enter the propensity model, and pairs match <i>exactly</i> on number of tags (Tables 2 and 3; largest residual SMD 0.041). Entering these controls in the Cox model leaves the pooled increment unchanged (1.04 versus 1.05) and strengthens the newcomer increment (1.13 versus 1.07; Tables G.14 and G.16). We report a caution alongside that gain rather than resting on it. Under the same controls the newcomer waiting-period diagnostic rises in step, from $\beta_2 = +0.050$ to $+0.122$ — essentially the magnitude of the arrival log-increment itself. Both terms inflate together, so we read the specification as ruling out observable selection as the <i>sole</i> driver, not as independent corroboration. The exception is a text-based complexity score: building a defensible one for 21 million question bodies was beyond what this revision could support, and Section 6.3 names it as a next step. On non-randomness, Section 6.5 now states the position directly — the increment “is identified under several assumptions, but one does the work” — and separates selection on observables, which the matching handles, from stable unobservables, which the difference-in-differences removes, from time-varying unobservables, which neither device removes.

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(Q3) “The response-time analysis should be modeled more flexibly. The results appear closer to an inverted-U pattern than to a simple ‘faster is stronger’ effect.” (Q5) “H3 should be rewritten or toned down. The strongest effect occurs in the 30–60 minute bin, not among the fastest responses. This does not cleanly support the original hypothesis.” (Q4) “Figure 4 is useful, but the text should treat it as evidence of a non-linear response-time pattern rather than as simple support for faster responses.”	Agreed on the specification, and we rewrote the hypothesis. Ten discrete response-time bins replace the continuous interaction, each fit as a separate Model A (Table 9, Figure 4). The pattern is an early sweet spot rather than a monotone gradient: near-null under 30 minutes, 1.19 [1.17, 1.21] at 30–60 minutes, modest thereafter. H3 previously asserted a strictly monotone relation; it now reads “<i>Timelier responses to a user’s question strengthen the generalized reciprocity effect.</i>” One correction the reviewer’s question surfaced: the window closes two days after the <i>answer</i> arrives, not at a fixed $T_Q + 2D$, so β_4 is identified in every bin including the slow ones. The earlier draft described this incorrectly (Section 4). We have not, however, toned the finding down to non-support, and we think that would misdescribe the evidence. The peak survives strongly post-treatment quality controls as the significant early-window maximum (1.03 [1.02, 1.04], $p < .001$; Table G.17). We therefore present H3 as support for an immediacy account while stating plainly, in both Results and Discussion, that the fastest bins do not fit it. Naming what does not fit seems more useful to a reader than a general softening that leaves the reservation unspecified.
(Q4) “Table 5 should report hazard ratios and confidence intervals for both specifications, not only for the main model.”	Agreed, and fixed. The bounds were computed but dropped by the table generator; Table 5 now reports the coefficient, standard error, and HR with 95% CI for β_4 in both columns. The one remaining em-dash is the Main+Speed bootstrap cell, since the pair bootstrap was run for Model A alone.
(Q4; see also Reviewer 2, P7) “The authors should translate hazard ratios into absolute effect sizes, such as additional answers per 1,000 treated questions.” (Q7) “The small effect sizes should be discussed more explicitly.”	Agreed. Table 10, now in the main text inside “Implications for Platform Design,” reports absolute rate differences, NNT with confidence bounds, and the reviewer’s own unit — additional helping events per 1,000 answered questions. Newcomers: 8.0 events per 1,000, NNT 125.0 (95% CI 103.2–157.8); pooled NNT 158.4. One correction in the process: these quantities must be referenced to the <i>control-arm</i> event rate, not the both-arms rate the previous version used. We also flag that the outcome is a recurrent-event rate, so NNT here is an approximate “answers needed per additional helping event.”

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(Q7) <i>“The results are from Stack Overflow, a highly specific technical Q&A community, and may not generalize to other online communities.”</i>	Agreed; addressed jointly with Reviewer 2’s external-validity points below. Section 6.5 now leads on Stack Overflow’s distinctiveness and drops the claim that generalized reciprocity “likely operates” elsewhere.
(Q8) <i>“The theory section should be shortened and more directly connected to the empirical design.”</i> <i>“Some implementation details currently in the appendix should be moved or summarized in the main text because they affect interpretation of the results.”</i>	Agreed. Theory is about one-sixth shorter (roughly 1,550 to 1,290 words) and now links forward to the design twice, at the end of the empirical-challenges subsection and after Hypothesis 2. The tenure-stratified adoption figure moved from the appendix into Results (Figure 6) with its interpretive paragraph. The legacy lifetime-contributions table was removed rather than promoted, for the reason given above.
(Q9) <i>Would you recommend language editing? — “Yes.”</i>	Agreed. A full language and consistency edit was applied after the substantive revisions were complete, covering grammatical errors, terminology standardization, and citation punctuation. The abstract is now self-contained, and the Highlights are aligned with it.

Response to Reviewer 2

Reviewer comment	Response and changes to the manuscript
(Overall) <i>“The problem is that the topic is out-dated in the era of GAI. [...] This makes the topic not interesting.”</i> (P9) <i>“The paper treats its conclusions as timeless, but they are, at best, a pre-generative-AI baseline. The authors must add a dedicated limitations paragraph explicitly stating that their findings are a historical snapshot, not a prescriptive guide for contemporary or future platform design.”</i> (P2) <i>“The authors overgeneralise findings to all knowledge-sharing platforms, ignoring that Stack Overflow is uniquely technical, anonymous, and status-driven and critically, its ecosystem has been structurally disrupted by generative AI since 2023.”</i>	We made the changes requested, and we also disagree with the premise, so we address both. On the changes: Section 3 drops the broad-generalization claim, now cites the platform-decline evidence (del Rio-Chanona et al. 2024; Burtch et al. 2024), and corrects the platform statistics. A dedicated Discussion subsection (Section 6.6) adopts the reviewer’s “contemporary or future” wording verbatim and reports that our own data register the decline. On the premise, we think the timing of the sample is the reason the study is timely rather than a reason it is not. A window that predates the steepest AI answer-substitution supplies a pre-substitution baseline for what a human answering ecosystem contributed to contributor recruitment while that ecosystem was still intact — a measurement that cannot be taken again once substitution is complete. We are careful about what that does and does not license. We make no forward claim about the effect’s magnitude or its survival under AI substitution; our own post-2020 newcomer estimate is null and thin, and we report it as such. What the record supplies is structural: it locates whatever recruitment leverage generalized reciprocity carries at the point of entry, shows it decaying to zero with tenure, and shows that shape holding across roughly a decade of the platform’s changing incentive regimes. For a field now facing contributor scarcity as language models absorb entry-level questions, that is a testable hypothesis about which lifecycle stage to instrument and a re-runnable design for detecting erosion of the human-answering pipeline. It is not a prediction and not a directive.

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(P3) “The analytical sample only includes users who remained active long enough to post a question and be observed. Users who posted a question, received no answer, and immediately churned are excluded. Since this exclusion is likely correlated with the treatment (no answer) and the outcome (no future helping), the estimated treatment effect [...] is probably downward-biased. The authors should conduct a sensitivity analysis using Heckman-type selection correction or, at minimum, explicitly bound the possible bias.”	The exclusion the reviewer describes does not occur in this design, so we begin by clarifying the sampling frame. The pipeline imposes no minimum-activity, minimum-tenure, or re-observation filter. Asked-but-never-answered users are <i>retained</i> as the control pool, and because each matched pair shares a fixed, pair-equalized observation window, a user who asks and then leaves contributes zero helping events over a fully observed window. Churn is part of the outcome rather than attrition from the sample. We did not implement a Heckman correction: no matching covariate provides a credible exclusion restriction, and the recurrent-event time-varying Cox is not natively Heckman-compatible. Instead we measured directly what the matching step does discard, and report the accounting in Appendix Appendix C. That exclusion is one of <i>common support</i>, not of activity or tenure: roughly half the unanswered pool has no on-support match, so the retained control pool is drawn from more readily answered tags than the discarded tail. Its direction is the opposite of the bias the comment posits. Because the discarded questions are the <i>less</i> answerable ones, the retained controls set a higher, not lower, comparison baseline, so the reported contrast is if anything conservative for the churn-prone population. Discarding is also not concentrated in the newcomer stratum that carries our central result. We treat the churn point itself as a clarification of the sampling frame rather than a limitation of the design, so we address it here rather than in the manuscript.

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(P4) “The outcome variable counts only ‘posting an answer’ as a helping event. On Stack Overflow, users frequently reciprocate through upvoting, editing, commenting, or accepting an answer. [These] behaviours require far less effort and are especially common among novices who lack expertise. By excluding these actions, the study systematically undercounts reciprocal behaviour and misclassifies many reciprocating users as non-responders. The authors should re-estimate the model using a composite helping indicator (including upvotes and comments) or justify why only answers capture the theoretically relevant construct, with reference to prior reciprocity literature.” (P1) “The operationalisation of ‘help’ is too narrow (answers only), excluding upvotes, edits, and comments — forms of low-cost reciprocity that newcomers often use.”	We did both things proposed, and the re-estimation runs against the concern rather than for it. The justification: a new “Why answers anchor the construct” paragraph grounds the outcome in the experimental reciprocity literature, where generalized reciprocity is operationalized as a <i>costly</i> prosocial contribution (Greiner & Levati 2005; Stanca 2009; Mujcic & Leibbrandt 2018), and where it is the willingness to bear that cost, not low-cost approval, that distinguishes reciprocity from signaling (Molm et al. 2007). The re-estimation: Table 6 reports every broader definition the data support — answers-only 1.049, comments-only 1.013, edits-only 1.163, answers+comments 1.024, answers+comments+edits 1.045. Adding comments <i>dilutes</i> the estimate rather than amplifying it, which is the opposite of the mechanical newcomer-undercount the concern predicts. Two of the named channels cannot enter the model, for distinct reasons. Accepting an answer is a <i>self-directed</i> act, and only treated users ever receive an answer to accept, so the matched control has no counterpart behavior and the difference-in-differences contrast is degenerate rather than informative. Up-votes are unobservable in principle: the public dump’s up-vote records carry no voter <code>UserId</code>, so the signal cannot be constructed for any user — re-parsing would not recover what was never released. Section 6.5 states what the outcome does and does not include, and follows the reviewer’s own logic to its conclusion: if newcomers do reciprocate disproportionately through unobserved channels, our estimates miss more of their reciprocity than of the veterans’, which makes the reported tenure gradient conservative.

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(P5) “The paper interprets the peak reciprocity at 30–60 minutes as a ‘re-engagement window’ effect. However, a simpler alternative is answer quality: extremely fast answers (<15 min) are often terse, link-only, or low-scoring, while 30–60-minute answers are typically more detailed, with code snippets and explanations, generating greater gratitude. The authors do not control for answer length, vote score, or whether the answer was accepted. Without testing this confound (e.g., by conditioning on answer quality metrics), the structural ‘session-dynamics’ interpretation remains speculative and should be toned down.”	Agreed that the confound needed testing, and we tested it. We re-extracted three observable quality signals — first-answer length, vote score, and acceptance — and added them one at a time (Table G.15). Answer length alone, the reviewer’s terse-versus-detailed variable and the signal least entangled with the asker’s own behavior, barely moves the estimate (1.038 to 1.032); vote score brings it to 1.010; the full specification pushes it below one (0.982). We report these as post-treatment bounds rather than estimates net of quality, because acceptance requires the asker to return to the site and is endogenous to the outcome itself. We also verified that the below-one result is not an artifact of score coding: signed-log (asinh) and ordinal-bin codings leave the full specification below one, with only the intermediate score-only step sensitive to coding (Table G.18). We use asinh rather than $\log(1 + \text{score})$ because roughly two million answers carry net-negative scores, for which the latter is undefined. The re-engagement reading is now presented as one conjecture among three, alongside answer quality and question characteristics (Section 6.3).
(P5, continued) “The striking inverted-U response-time effect (peak at 30–60 minutes) conflates answer speed with answer quality; faster answers may be terse or incorrect, while 30-minute answers tend to be more thorough, a confound not tested.”	We re-estimated each of the ten bins with the same three controls (Table G.17). The 30–60-minute peak attenuates from 1.19 [1.17, 1.21] to 1.03 [1.02, 1.04] but remains the clearly significant ($p < .001$) early-window maximum, while the same controls push the fastest bins significantly <i>below</i> one (0.95). That below-one result is a signature of over-adjustment rather than a finding: acceptance requires the asker’s return, which is part of the very pathway that produces the outcome. Because the peak survives an adjustment biased against it, we retain the timing reading rather than replacing it with a quality reading. The secondary elevation at one to three days (1.09) is left uninterpreted for the same reason.

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(P6) <i>“The platform has undergone massive cultural, technical, and policy shifts from 2008 to 2025 — including reputation inflation, spam filters, tag evolution, and most critically, the rise of ChatGPT post-2022. [...] the tenure-stratified analysis pools newcomers from 2008 with those from 2024 as if they share the same ‘newcomer’ experience. This ignores cohort heterogeneity. The authors should include year-of-entry fixed effects or conduct robustness checks on pre-2020 subsamples to isolate the pure human-only reciprocity mechanism.”</i>	We took the second of the two options, because the first is not estimable here: year-of-entry fixed effects would be collinear with the design, since calendar year already enters as an exact-matching stratum and registration cohort is absorbed once question-year is matched. Re-estimating Model A before 2020 leaves the arrival increment essentially unchanged (Table G.13). That subsample retains 79% of matched rows and 91% of helping events, however, so we describe it as a compositional check rather than an independent replication — and we went beyond the request by estimating the <i>post-2020</i> window separately, which shares no rows with it. We report both outcomes, including the less convenient one. The pooled increment replicates across eras (post-2020 HR 1.04, [1.03, 1.06]). The newcomer estimate is not distinguishable from one after 2020 (1.00, [0.97, 1.03]) and rests on far fewer events (31,814 versus 336,966), which is consistent with the post-2022 contraction the reviewer describes. Section 5 therefore states that the newcomer association is identified in the pre- and early-generative-AI data that dominate the sample, and Section 6.6 offers two conjectures for the attenuation — cohort selection among post-2022 joiners, and a residual question mix skewing beyond a newcomer’s ability — both marked untestable with archival data.
(P7) <i>“The paper does not report the Number Needed to Treat (NNT) — i.e., how many newcomers must receive an answer to generate one additional helping event. Without this, platform designers cannot assess whether investing in faster response systems yields meaningful returns. The authors should calculate and report NNT, and revise their ‘Implications for Design’ section to reflect modest effect sizes.”</i>	Agreed; addressed jointly with Reviewer 1’s absolute-effects request above. Table 10 reports NNT with confidence bounds and events per 1,000 answered questions by tenure bucket, placed in the main text beside the design claims it disciplines, and “Implications for Platform Design” declines to recommend interventions built specifically on response speed.

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(P8) <i>“Posting a question increases a user’s profile visibility: other members click on their profile, view their history, and may trigger a sense of social obligation or reputation concern. This ‘exposure effect’ is not equivalent to gratitude for receiving an answer, yet it operates precisely during the post-answer period and is not separately identified. The waiting-period covariate absorbs baseline question-asking effects, but it does not capture the increased visibility that persists after an answer arrives. A placebo test using questions that received no answers but high views could help rule out this confound; currently, it is not attempted.”</i>	We agree this bears on mechanism <i>attribution</i>, and the channel now appears in the manuscript under a name — <i>social expectation</i>. Because the exposure is caused by the answer, we treat it as a candidate mechanism for the behavior change rather than as a threat to identification: it is now a third proposed pathway in the Theory section, alongside gratitude and moral obligation, and Section 6.5 states plainly that our design cannot say which of the three operates and leaves that question to psychologists with the instruments to measure it. We would resist the assumption that the channel runs in the direction the comment supposes. The exposure must be carried by contemporaneous, logged-in users who browse question feeds — the same population that supplies potential answerers — and for them, being answered cuts both ways: it lifts a question in activity-sorted feeds, but it also takes the question out of the “unanswered” queue those users browse and marks it resolved. Which component dominates we cannot establish, so the manuscript claims no more than that the change in visibility is not obvious (Section 6.5). We would add one observation about the tenure gradient specifically, which we make here rather than in the paper: for social expectation to produce it, being watched would have to bind newcomers more tightly than veterans, yet veterans have more reputation at stake and more visible profiles and do not respond at all. The proposed placebo is not identified in these data. The dump records <i>ViewCount</i> only as a single cumulative snapshot at the collection date, dominated by later, largely logged-out search-engine visitors who never reach a profile — the wrong population for the construct — so it conflates a question’s age, its topic’s popularity, and years of downstream traffic with the contemporaneous attention the placebo would require. Testing the channel directly needs time-stamped view data or repeated snapshots, which do not exist for our study period. A related channel, asker re-engagement, likewise coincides with answer receipt. We guard against its most obvious form by defining treatment as answer <i>receipt</i> rather than acceptance, and treat the rest as a conjecture we cannot identify (Section 6.3).
(Q9) <i>Would you recommend language editing? — “Yes.”</i>	Addressed jointly with Reviewer 1’s Q9 above.

We hope these revisions address the reviewers' concerns. We are grateful for feedback that materially improved the identification, the transparency of the sample and inference, and the calibration of our claims.